Week 9 – HCI vs HRI Safety Considerations

Student Name: Ansarah Mohammed

Programme: CariSurg MedTech Pathways

Week: Week 9 – Designing & Prototyping Human-Centred Systems (HCI/HRI)

Chosen Implementation: AI-Assisted Emergency Department Triage Support System (HCI)

HCI-Specific Safety Considerations:

Safety Concern	Why it Matters	Mitigation
Alarm Fatigue	During busy Emergency Department shifts, frequent notifications may cause nurses to overlook or delay responding to genuinely urgent alerts.	Prioritise alerts by clinical severity, minimise unnecessary notifications, allow configurable thresholds and regularly review alert performance.
Display Legibility Under Stress	Nurses often work under time pressure, fatigue and frequent interruptions. Poor interface design may delay recognition of important patient information.	Use high-contrast colours, large readable text, consistent layouts, and combine colour with icons and labels so information remains accessible under pressure.
Automation Bias	Clinicians may become overly reliant on AI recommendations and unintentionally reduce independent clinical judgement.	Display confidence scores and contributing clinical factors, require manual confirmation or override, and reinforce that the AI is a decision-support tool rather than the final decision-maker.

HRI-Specific Safety Considerations:

Safety Concern	Why it Matters	Mitigation
Sensor Reliability and Failure	Incorrect or incomplete vital-sign measurements may lead to inaccurate observations or delayed recognition of patient deterioration.	Validate sensor readings, detect incomplete measurements, prompt repeat observations where necessary, and immediately notify staff if the assessment cannot be completed.
Accessibility and Digital Literacy	Observation Units may serve patients with varying levels of digital confidence, particularly older adults or individuals with limited experience using digital health technologies. Some patients may also have visual, hearing or mobility impairments.	Design the kiosk with large buttons, simple language, multilingual support where appropriate, clear step-by-step instructions, and an easy way to request assistance from a healthcare professional at any time.
Physical Interaction and Patient Safety	The kiosk operates in a clinical environment where patients may have reduced mobility or require assistance. Poor placement or movement could create safety risks or discourage use.	Use a stable, stationary kiosk with clearly marked operating areas, minimise moving components, maintain adequate space for mobility aids such as wheelchairs, and ensure staff supervision is available whenever required.

Within Caribbean healthcare settings, hospitals often care for a diverse patient population with varying levels of digital literacy, health literacy and access to technology. Older adults and first-time users may require additional guidance when interacting with digital systems, while hospitals may also experience resource limitations, intermittent network connectivity or staffing constraints. Human-centred design should therefore prioritise simplicity, accessibility, clear communication and reliable manual fallback procedures to ensure the system remains safe and usable for all patients.